

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 38)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

For each of the following investigations, name the most appropriate statistical test.

[4 marks]

Whether noise level decreases as distance from a road increases

.....

Whether the numbers of dormice differ between woodlands with different management ..

.....

Whether mean dissolved oxygen differs between two streams

.....

Whether crop yield increases with fertiliser application (ranked data)

.....

0 1 . 2

Explain why the chi-squared test cannot be used for data such as wind speed or plant height.

[1 mark]

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.....

0 1 . 3

A Spearman's rank test on 12 pairs of data gave $r_s = -0.72$. The critical value at $p = 0.05$ for $n = 12$ is 0.59. Explain what can be concluded.

[3 marks]

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0 1 . 4

Explain the purpose of error bars on a graph of mean values.

[2 marks]

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10

0 2 . 1

Explain why carnivorous species such as salmon dominate aquaculture in MEDCs while herbivores such as carp and tilapia dominate in LEDCs.

[3 marks]

0 2 . 2

Describe how all-female rainbow trout are produced and explain why this is done.

[3 marks]

0 2 . 3

Explain how each of the following abiotic factors is controlled in an intensive salmon farm.

[4 marks]

Dissolved oxygen

.....

Water flow

.....

Day length

.....

Temperature

.....

10

03 . 1

Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

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03 . 2

Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

1

2

3

4

0 3 . 3 Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each. **[3 marks]**

1

2

3

10

0 4 Table 1 shows the number of plant species per m² in a chalk grassland reserve under different management, ten years after management began.

Table 1

Management	Mean plant species per m ²	Percentage of area covered by scrub / bushes
No management (abandoned)	8	55
Sheep grazing	24	5
Annual mowing (August)	19	0
Winter burning	14	10

0 4 . 1 Explain what is meant by a **plagioclimax** community and why chalk grassland is an example. **[2 marks]**

0 4 . 2

Use Table 1 to explain why grazing produced the highest plant diversity.

[4 marks]

0 4 . 3

Explain why different grazing animals are chosen for different plagioclimax habitats.

[2 marks]

0 4 . 4

Explain how burning maintains heathland.

[2 marks]

10

0 5

The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

0 5

. 1

Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

1

2

3

4

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0 5

. 2

Explain why involving local communities is essential to the success of the reserve.

[3 marks]

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0 5

. 3

Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

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10

06 . 1 Describe how each of the following threatens Antarctica.

[4 marks]

Global climate change

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Ozone depletion

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Tourism

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Future mineral exploitation

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06 . 2 Describe the main features of the Antarctic Treaty (1959) and the regulations that protect Antarctica.

[6 marks]

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10

0 7 Biomimetics uses knowledge of the adaptations of other species to improve the design of manufactured items.

0 7 . **1** Complete Table 1 by adding the missing adaptation or application.

[4 marks]

Table 1

Species and adaptation	Design application	
Tubercles (bumps) on the flippers of humpback whales		
	Self-cleaning glass	
Shark skin scales that reduce friction		
	Velcro	

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0 7 . **2** Explain how the design of termite mounds has been copied to reduce the environmental impact of large buildings.

[3 marks]

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0 7 . 3

Describe **two** ways in which studying the physiology of other species could help to solve human health problems.

[2 marks]

1

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2

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0 7 . 4

Explain why bird wing bones are a useful model for engineers.

[1 mark]

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10

0 8 . 1

State what is meant by **hybrid vigour**.

[1 mark]

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0 8 . 2

Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 8 . 3

Give **two** potential applications of cloning to livestock farming.

[2 marks]

1

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2

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5

09 . 1

Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

[6 marks]

Water availability

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Dissolved oxygen

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pH

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09 . 2

Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

[2 marks]

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09 . 3

Explain why species that were already rare are most likely to be lost when abiotic conditions change.

[2 marks]

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10

10 . 1

Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

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Extra space

10 . **2** State what is meant by a **taxon**.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

OR

1 1 . 2

Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

Do not write outside the box

END OF QUESTIONS